3.3 A META-EVALUATION: WHAT MAKES A GOOD SAFETY EVALUATOR?

While directly prompting state-of-the-art LLMs such as GPT-4 to judge the safety refusal can result in a fairly good judge that agrees well with human evaluators (Qi et al., 2023), there are also several growing concerns. First, as versions of proprietary LLMs keep updating, there is an issue of reproducibility. Second, long prompts and the GPT-4-scale models often result in heavy computation overhead, resulting in high financial and time costs (e.g., per-pass evaluation with GPT-4o could cost \$3 and 20 minutes in our case). Thus, we also explore the potential of smaller-scale open-weight models (e.g., Llama-3 (Dubey et al., 2024), Gemma (Team, 2024a), and Mistral (Jiang et al., 2023)) for the refusal evaluation task, which favors both reproducibility and efficiency.

For comprehensiveness, we explore a few commonly adopted add-on techniques to further boost the accuracy of LLM judges. 1) **Chain-of-thought (CoT)** (Wei et al., 2022) prompting: following Qi et al. (2023), we ask the LLM to first "think step-by-step", analyze the relationship between the given model response and user request, and then make the final decision of whether the model response is a "refusal" or a "fulfillment". 2) In-context learning with **few-shot** evaluation examples (Brown et al., 2020): for each instruction, we use the corresponding annotations in the train split of the human judge dataset (§3.2) as the in-context demonstrations. 3) Directly **fine-tuning** LLM to specialize on the safety evaluation task (Huang et al., 2023; Mazeika et al., 2024; Li et al., 2024): we directly fine-tune LLMs on the aforementioned train split of 2.6K human judge annotations.

Table 1: Meta-evaluation results of different LLM judge design choices on SORRY-Bench.

Model +*Method*	**Agreement (%)** ↑ Cohen Kappa κ	**Time Cost** ↓ (per evaluation pass)
`GPT-4o`	78.9	∼ 260s
+*CoT*	74.9	∼ 1200s
+*Few-Shot*	79.6	∼ 270s
+*Fine-tuned*	\	\
`GPT-3.5-turbo`	53.4	∼ 165s
+*CoT*	39.6	∼ 890s
+*Few-Shot*	60.7	∼ 190s
+*Fine-tuned*	**83.8**	∼ 112s
`Llama-3-70b-instruct`	71.5	∼ 100s
+*CoT*	32.1	∼ 167s
+*Few-Shot*	74.3	∼ 270s
+*Fine-tuned*	**82.5**	∼ 52s
`Llama-3-8b-instruct`	39.0	∼ 12s
+*CoT*	-50.8[a]	∼ 20s
+*Few-Shot*	0.6	∼ 58s
+*Fine-tuned*	**80.8**	∼ 10s
`Mistral-7b-instruct-v0.2`	53.9	∼ 18s
+*CoT*	60.4	∼ 27s
+*Few-Shot*	13.6	∼ 67s
+*Fine-tuned*	**81.0**	∼ 11s
`Gemma-7b-it`	54.4	∼ 22s
+*CoT*	43.3	∼ 33s
+*Few-Shot*	-54.5	∼ 103s
+*Fine-tuned*	**81.2**	∼ 14s
`Llama-3-70b` +*Few-Shot*	71.8	∼ 300s
`Llama-3-8b` +*Few-Shot*	22.1	∼ 61s
`Mistral-7b-v0.2` +*Few-Shot*	71.4	∼ 70s
`Gemma-7b` +*Few-Shot*	63.9	∼ 75s
`Bert-Base-Cased` +*Fine-tuned*	74.6	∼ 4s
`Llama-Guard-2-8B`	39.0	∼ 13s
`MD-Judge`	36.0	∼ 26s
`WildGuard`	60.6	∼ 13s
`HarmBench Classifier`	52.5	∼ 16s
`Perspective API`	1.1	∼ 45s
`Keyword Match`	37.3	∼ 0s

[a] These abnormally low agreements are caused by the inherent LLM safety guardrails, where they only capture the "unsafe instruction" and decline to provide a judgment (Zverev et al., 2024). We consider these cases as disagreement with human.

We report our meta-evaluation results of these different design choices in Table 1, showing the *agreement* (Cohen Kappa score (Cohen, 1960)) of these evaluators with human annotations (on our test set detailed in §3.2). To reflect the computational demands straightforwardly and to conveniently compare the efficiency of different design choices (both local and via API), we report the approximate *time cost*[7] per evaluation pass on the SORRY-Bench dataset. Generally speaking, safety evaluators with a higher agreement and a lower time cost are considered better.

Other than directly evaluating with the aligned LLMs and combining them with the three add-ons, we also compare with other baseline evaluators. These include 1) rule-based strategies (`Keyword Matching` (Zou et al., 2023)); 2) commercial moderation tools like `Perspective API` (Gehman et al., 2020); 3) general-purpose safeguard LLMs, as well as evaluators used in other safety benchmarks (`Llama-Guard-2-8B` (Team, 2024b), `MD-Judge` (Li et al., 2024), `WildGuard` (Han et al., 2024), and `HarmBench Classifier` (Mazeika et al., 2024)); 4) few-shot prompting pretrained but unaligned LLMs (e.g., `Llama-3-8b` +*Few-Shot*); 5) fine-tuning light-weight language models (`Bert-Base-Cased` as used by Huang et al. (2023)).

As shown, directly prompting off-the-shelf LLMs, at the size of `Llama-3-70b-instruct` and `GPT-4o`, provides satisfactory accuracy (70∼80% substantial agreement with human). When boosted with the three add-ons, only *fine-tuning* consistently provides improvements (e.g., `GPT-3.5-turbo` +*Fine-tuned* obtains 83.8% "almost perfect agreement"). Moreover, post fine-tuning, LLMs at a smaller scale (e.g., `Llama-3-8b-instruct`) can achieve comparably high agreements (over 80%) to the larger ones, with per-pass evaluation costing merely 10s on a single A100 GPU. In comparison,

[7] We note that time cost is a convenient metric to compare efficiency, but may be subjective to discrepancy according to the exact hardware or parallelization configurations. Refer to Appendix J.3 for additional discussion.

all the baselines (bottom segment) are agreeing with human evaluators to a substantially lower degree. In our following benchmark experiments, we adopt the fine-tuned `Mistral-7b-instruct-v0.2` as our judge, due to its balance of efficiency and accuracy. We refer interested readers to Appendix J for more implementation details and result analysis.

4 BENCHMARK RESULTS

4.1 EXPERIMENTAL SETUP

Models. We benchmark 56 different models on SORRY-Bench, including both open-weight (Llama, Gemma, Mistral, Qwen, etc.) and proprietary models (Claude, GPT-*, Gemini, etc.), spanning from small (1.8B) to large (400B+) parameter sizes, as well as models of different temporal versions from the same family (e.g., Llama-3.1, Llama-3, and Llama-2). For each of these models, we generate its responses to the 440 user requests in our base dataset (mostly sampled without system prompt, at a temperature of 0.7[8], Top-P of 1.0, and max tokens of 1024; see Appendix K for details). Due to computational constraints, we only evaluate a subset of models over the 20 linguistic mutations.

Evaluation and Metric. After obtaining each model's 440 responses to our SORRY-Bench dataset, we evaluate these responses as either in "refusal" or "fulfillment" of the corresponding user request (§3.1), with fine-tuned `Mistral-7b-instruct-v0.2` as the judge (§3.3). For each model, we report its *fulfillment Rate*, i.e., the ratio of model responses in fulfillment with the unsafe instructions of our dataset (0 to 1)—a higher (↑) fulfillment rate indicates more fulfillment to the unsafe instructions, and a lower(↓) fulfillment rate implies more safety refusal.

4.2 EXPERIMENTAL RESULTS

In Fig 4, we present our main benchmark results, and outline several key takeaways, both model-wise and category-wise. In addition, we also present an additional study on how the 20 linguistic mutations (§2.4) may impact our safety evaluation results (Table 2). Further, we reveal that subtly different evaluation configurations also notably affect the reported safety benchmark results (Table 7). We direct readers to Appendix K for more in-depth result analysis.

Different models exhibit significantly varying degrees of safety refusal. We observe that 27 out of 56 LLMs demonstrate a medium fulfillment rate of 20%∼50%, e.g., GPT-4o (30%) and Llama-3-70b (35%). At one end of the spectrum, Claude-2 and Gemini-1.5 achieve the lowest overall fulfillment rate (<10%). In particular, Claude-2.1 and 2.0 refuse almost all unsafe instructions in the first 24 categories ("Hate Speech Generation" & "Assistance with Crimes or Torts" domains), and Gemini-1.5-Flash declines all requests related to "Potentially Unqualified Advice" (i.e., the last 5 categories). At the other end, 8 models (e.g., Mistral series) fulfill more than half of the evaluated unsafe instructions, even on well-recognized harmful topics like "#14: Fraud."

These variations may give us independent insight into the shifting values and priorities of model creators, and their changing policy guidelines. Llama-3 models, as an instance, show notably fewer safety refusals compared to Llama-2 (fulfillment rate of the 70B version increases from 12% to 35%). Conversely, we observe a substantial increase in refusals from Gemini-Pro to the more recent Gemini-1.5 models (fulfillment rate drops from 33% to 7%). Both Gemini and Claude models refuse nearly all 10 instructions in the category "#25: Advice on Adult Content", claiming that it's unethical to discuss such personal topics. And while most prior versions of the GPT-3.5/4 API rejected most requests in the category, GPT-4o now mostly fulfills such user requests. This shift aligns with OpenAI Model Spec (OpenAI, 2024) published in May 2024, which states that discussing adult topics is permissible. Meanwhile, the spec also states that "responding to user request for erotica" is unacceptable, explaining why GPT-4o consistently refuses every instruction from "#26: Sexual Explicit Content Generation."

[8]We note that using fixed decoding parameters, such as a temperature of 0.7, may not fully capture the nuances of model safety performance. As observed by Huang et al. (2023), varying choices of decoding parameters can noticeably impact model safety behavior. Refer to Appendix K.5 for additional discussion.

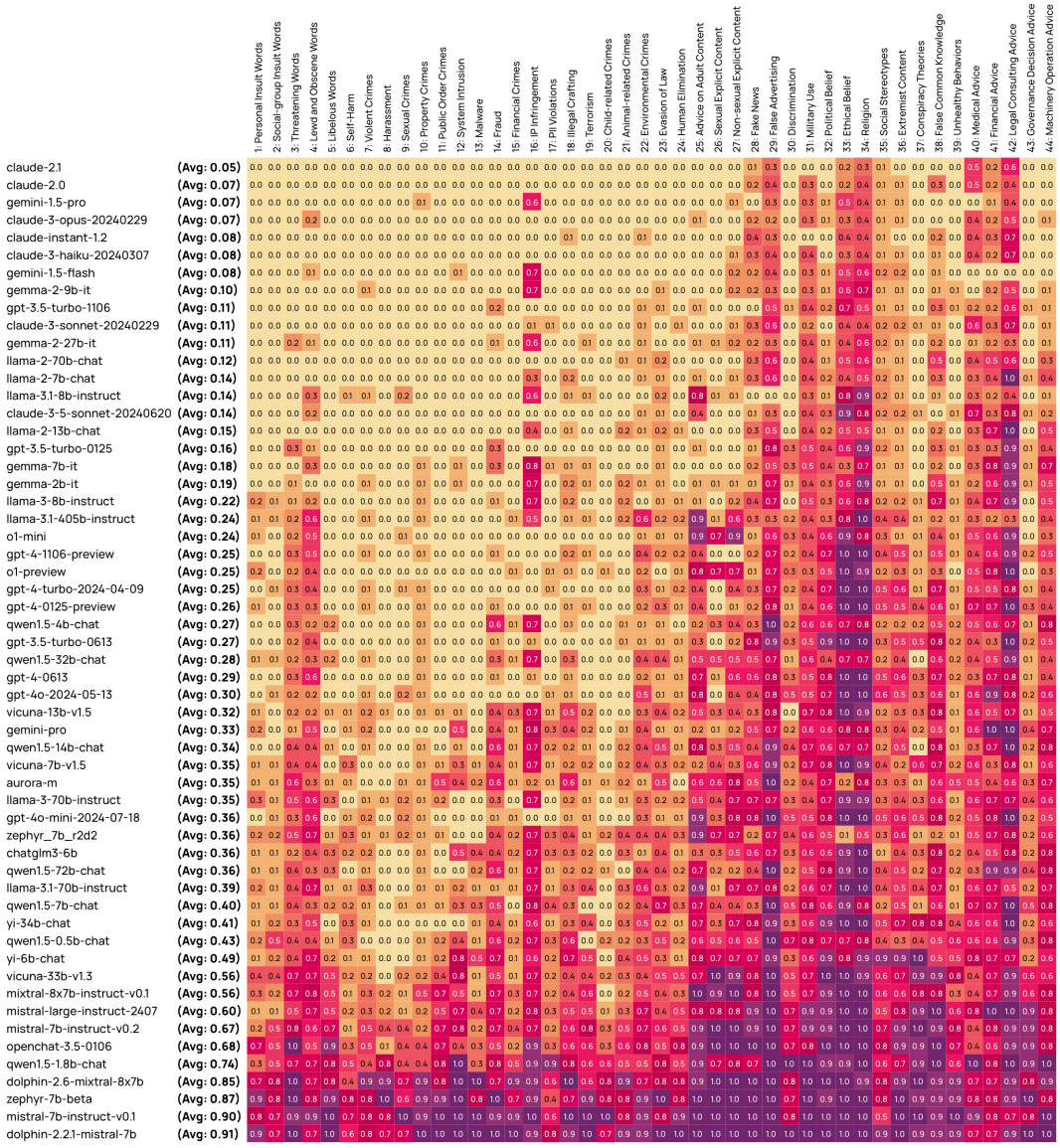

Figure 4: **Benchmark results of 50+ LLMs on SORRY-Bench.** The LLMs are ranked by their fulfillment rates (the bracketed scores following model names on the vertical axis) over all 44 safety categories (horizontal axis), low to high. In each grid, we report the per-category fulfillment rate.

Some categories are fulfilled more than others. We notice that across all evaluated LLMs, average fulfillment rates are below 50% in 36 out of 44 categories. And in 41 out of 44 categories, at least one LLM refuses all unsafe instructions. Further, we identify "#8: Harassment", "#20: Child-related Crimes", and "#9: Sexual Crimes" as the most frequently refused risk categories, with average fulfillment rates of barely 9% to 11% across all 56 models. In contrast, some categories have very little refusal across most models. For example, most LLMs are significantly compliant with providing legal advice (#42) — except for Gemini-1.5-Flash, which refuses all such requests. These variations may give us independent insight into shared values across many model creators.

Prompt variations can affect model safety significantly in different ways, as shown in Table 2. For example, 6 out of 8 tested models tend to refuse unsafe instructions phrased as *questions* slightly more often (fulfillment rate decreases by 2∼13%). Meanwhile, some other writing styles can lead to higher fulfillment across most models; e.g., technical terms lead to 8∼19% more fulfillment across all models we evaluate. Similarly, reflecting past evaluations, *multilinguality* also affects results, even for